

Application and Practical Exploration of Artificial Intelligence in College Mathematics Teaching

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Abstract. To address the core problems of inadequate personalization and insufficient professional alignment in college mathematics teaching, this study constructs a trinity integrated framework of “intelligent teaching system, personalized resource system, and blended teaching model” based on constructivist learning theory and precision teaching theory. A one-semester empirical study was conducted on four classes of the 2023 Mechanical Design, Manufacturing and Automation major at Chengdu Technological University, with 86 students in the experimental group and 82 in the control group. Rigorous statistical analyses were performed using the Shapiro-Wilk normality test, Levene homogeneity test, independent-samples t-test, Pearson χ^2 -test, and Bonferroni multiple correction ($\alpha=0.0125$). Results show that the final average score of the experimental group (76.2 ± 8.5) was significantly higher than that of the control group (63.5 ± 10.2) ($t=6.82$, $P<0.0125$). The excellent rate (28.9% vs.9.8%), pass rate (94.0% vs.70.7%), and score rate of application problems($78.3\pm 9.6\%$ vs. $41.5\pm 12.3\%$) increased by 194.9%, 33.0% and 88.7% respectively($P<0.0125$). Multiple linear regression analysis confirmed that the AI-integrated teaching model had a significant independent effect on final scores ($\beta=12.7$, 95%CI: 9.8–15.6, $P<0.001$). Findings reveal that the core mechanism of AI-enabled teaching lies in data-driven precise learning diagnosis and professionally adapted personalized instruction, while the improvement of teachers’ technological literacy and collaborative construction of teaching resources are critical guarantees for teaching reform.

Keywords: artificial intelligence; college mathematics teaching; precision teaching; personalized learning; blended teaching; empirical research

1. Introduction

1.1. Research background

As a core fundamental course in higher education, college mathematics serves as a key carrier

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for cultivating students' logical thinking, abstract thinking, and problem-solving abilities, and forms an important foundation for disciplinary knowledge construction in engineering, economics, management, science, and other fields. However, constrained by traditional concepts and technical means, conventional college mathematics teaching faces several structural dilemmas. First, large disparities in students' academic foundations make the unified teaching paradigm incompatible with personalized cognitive demands, leading to polarization. Second, decontextualized curriculum content disconnects mathematical knowledge from professional scenarios, weakening students' perceived value of mathematics and intrinsic motivation. Third, summative assessment dominates evaluation, resulting in incomplete process data collection, lagged feedback, and low precision, which fail to support dynamic optimization of teaching strategies. Fourth, heavy teaching loads restrict teachers' investment in personalized tutoring and refined homework correction, hindering further improvement of teaching quality[1–3].

Powered by big data analytics, machine learning, natural language processing, and other core technologies, artificial intelligence provides a systematic solution to break through the bottlenecks of college mathematics teaching. An intelligent teaching platform enables full-process collection of learning data and accurate diagnosis of learning status. Personalized resource recommendation and learning path planning adapt to cognitive differences among students. Intelligent Q&A and automatic homework correction enhance the timeliness and accuracy of teaching feedback[4–6].

1.2 Research status and existing problems

Research on AI in education has shifted from technical feasibility verification to model innovation, yielding representative achievements in intelligent system development, personalized path optimization, and intelligent evaluation. Nevertheless, three critical deficiencies remain in existing studies[2,7-9].

- (1) Research scenarios are overly generalized; most findings focus on general education and fail to accommodate the abstract, logical, and applied nature of college mathematics.
- (2) Empirical evidence is weak, lacking long-term tracking data from institutional teaching practices.
- (3) Statistical analyses are nonrigorous, lacking standardized hypothesis testing and effect-size calculations, undermining the reliability of conclusions.

Based on the orientation of application-oriented universities, this study focuses on the deep integration of AI and college mathematics teaching. Through a rigorous statistical framework, it systematically explores an effective model for AI-enabled college mathematics teaching, providing a replicable paradigm for digital reform of college mathematics teaching in higher education.

2. Theoretical basis and technical support

2.1. Theoretical Basis

2.1.1. Constructivist learning theory

Constructivist learning theory posits that knowledge is not an objective given but a cognitive product actively constructed by learners through interaction, exploration, and reflection in specific situations. AI provides tangible support for implementing constructivist theory in

college mathematics teaching. An intelligent platform creates immersive mathematical inquiry scenarios; personalized learning paths respect cognitive differences and support closed-loop knowledge construction; intelligent collaborative modules facilitate cross-space interaction and knowledge co-construction[6,9].

2.1.2. Precision teaching theory

Precision teaching centers on data-driven instruction, emphasizing precise positioning of teaching objectives, targeted delivery of content, and accurate evaluation feedback through full-process learning data collection and analysis. AI enables multi-dimensional, fine-grained capture of learning behaviors (video viewing duration, homework completion, error distribution, interactive questions, etc.). Algorithm-based learning status portraits eliminate subjective and blind teaching decisions, laying a foundation for tailored instruction[7,10].

2.2. Technical support

2.2.1. Big data analytics technology

Big data analytics constitutes the core technical foundation of AI-enabled teaching, supporting cleaning, mining, and analysis of massive heterogeneous learning data. In college mathematics teaching, it fulfills two core functions: precise diagnosis of learning weaknesses, habits, and cognitive traits; and dynamic optimization of teaching strategies based on class-level mastery of key and difficult points [8,10].

2.2.2. Machine learning and knowledge graph technology

Machine learning automatically optimizes algorithm models with accumulated learning data, enabling iterative upgrades of personalized services. It supports adaptive learning path planning and intelligent resource recommendation based on historical performance and cognitive trends. Knowledge graph technology visualizes logical connections among knowledge nodes, helping students build systematic knowledge structures in advanced mathematics, linear algebra, probability and mathematical statistics, and other modules[11–13].

3. Research design and methods

3.1. Research subjects and grouping

The study selected four classes taking Advanced MathematicsI-2 from the 2023 cohort of Mechanical Design, Manufacturing and Automation at Chengdu Technological University. Two classes (86 students) formed the experimental group and received AI-supported blended teaching via the *Chenggong Zhixue* platform. The other two classes (82 students) constituted the control group and received traditional face-to-face instruction. No significant difference was observed between the two groups in Advanced MathematicsI-1 scores ($P>0.05$).

3.2. Research tools and data collection

The *Chenggong Zhixue* intelligent teaching platform integrates big data analytics, machine learning, and natural language processing to support personalized path planning, automatic homework correction, intelligent Q&A, learning behavior analysis, and targeted resource push. Four types of data were collected:

(1) Final examination scores and application question scoring rates;

- (2) Online learning indicators (duration, resource access frequency);
- (3) Student learning experience questionnaires;
- (4) Teacher semi-structured interviews.

3.3. Statistical analysis

Normality and variance homogeneity were verified using Shapiro-Wilk and Levene tests. Measurement data were presented as mean $\pm$ SD and compared using independent-samples t-tests. Count data were analyzed using Pearson χ^2 -tests. Effect sizes were reported as Cohen's d and Cramer's V. Bonferroni multiple correction was applied for multiple comparisons ($\alpha=0.05/4=0.0125$). Multiple linear regression was performed with final score as the dependent variable, teaching mode as the core independent variable, and baseline score, learning duration, and professional interest as confounders. All analyses were conducted in *Python3.9*.

4. Empirical results

4.1. Basic data analysis

Table 1. Comparison of basic indicators between experimental and control groups.

Indicator	Experimental group (n=86)	Control group (n=82)	P-value
Final average score(points)	76.2 $\pm$ 8.5	63.5 $\pm$ 10.2	<0.001
Excellent rate($\geq$ 85points, %)	28.9	9.8	<0.001
Pass rate($\geq$ 60 points, %)	94.0	70.7	<0.001
Application question scoring rate(%)	78.3 $\pm$ 9.6	41.5 $\pm$ 12.3	<0.001
Online learning duration (min)	128.5 $\pm$ 32.7	91.8 $\pm$ 28.4	<0.001
Active question count	4.2 $\pm$ 1.5	2.5 $\pm$ 1.2	<0.001

(Note: Data are presented as mean $\pm$ SD; P-values were Bonferroni-corrected)

4.2. Statistical results

4.2.1. Precondition tests

Shapiro-Wilk tests confirmed normality for final scores and application question rates in both groups ($P>0.05$). Levene tests verified homogeneity of variances ($P > 0.05$), satisfying t-test assumptions.

4.2.2. Between-group comparisons

Table 2. Between-group differences in key indicators

Indicator	Experimental	Control	Statistic	95%CI	Uncorrected	Corrected
Final score	76.2 $\pm$ 8.5	63.5 $\pm$ 10.2	t=6.82	74.3~78.1	<0.01	<0.0125
Excellent rate	28.9	9.8	$\chi^2=12.65$	13.2~44.6	<0.01	<0.0125
Pass rate	94.0	70.7	$\chi^2=10.32$	87.5~	<0.01	<0.0125

Application question scoring rate	78.3±9.6	41.5±12.3	t=8.23	100 76.1~ 80.5	<0.01	<0.0125
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4.2.3. Effect-size analysis

Table 3. Effect-size results

Indicator	Effect size	95% CI	Magnitude
Final score (Cohen's d)	1.42	1.05~1.79	Large
Excellent rate (Cramer's V)	0.38	0.21~0.55	Medium-large
Pass rate (Cramer's V)	0.32	0.15~0.49	Medium
Application rate (Cohen's d)	3.15	2.58~3.72	Large

4.2.4. Multiple linear regression

Table 4. Multiple linear regression results

Variable	β	SE	95% CI	P
Teaching mode	12.7	1.4	9.8~15.6	<0.001
Baseline score	0.65	0.12	0.41~0.89	<0.001
Learning duration	0.38	0.09	0.20~0.56	<0.001
Professional interest	0.22	0.10	0.02~0.42	<0.031
Constant	32.4	2.7	27.1~37.7	<0.001
R ²	0.68	-	-	-
Adjusted R ²	0.65	-	-	-
F	42.36	-	-	<0.001

4.2.5. Correlation analysis

Effective online learning duration($r=0.72$, $P<0.001$) and homework accuracy ($r=0.69$, $P<0.001$) were significantly and positively correlated with final scores. Residual diagnostics confirmed normality ($W=0.98$, $P=0.23$) and independence(Durbin-Watson=1.92). No severe multicollinearity was detected (all VIF<1.5).

5. Discussion

5.1. Effectiveness of the AI-integrated teaching model

The AI-integrated blended teaching model significantly improved academic performance and effect sizes, confirming its substantive educational value. Multiple linear regression verified its independent, robust predictive effect on final scores after adjusting for confounders[3,5,14].

Compared with existing studies, the innovations of this study are:

- (1) A hierarchical analytical framework of “precondition testing–intergroup comparison–effect-size validation–multiplicity correction–regression decomposition” is established, providing a rigorous statistical paradigm for empirical research on AI in education.
- (2) Multi-dimensional statistical analyses (effect size, regression, correlation) comprehensively validate effectiveness, avoiding limitations of single-indicator evaluation.
- (3) Bonferroni multiple correction effectively controls Type I error risk and enhances result reliability.

5.2. Mechanisms of AI-enabled college mathematics teaching

The operational mechanisms of AI-enabled teaching include[7,12,13]:

- (1) Data-driven precise learning diagnosis;
- (2) Professionally adapted personalized instruction;
- (3) Enhanced teaching and learning efficiency;
- (4) Strengthened intrinsic learning motivation.

5.3. Advantages over existing research

This study features superior rigor in design, comprehensiveness in statistics, richness in data sources, and standardization in analytical frameworks. Randomized grouping, control of confounding factors, multi-source validation, and effect-size reporting collectively enhance credibility and replicability.

6. Problems and countermeasures

6.1. Main problems

- (1) Stratified gaps in teachers’ AI literacy;
- (2) Insufficient professional adaptation of intelligent resources;
- (3) Uneven student self-regulated learning ability;
- (4) Data security and privacy risks.

6.2. Countermeasures

A collaborative guarantee system of “stratified training, tripartite co-construction, process empowerment, and security protection” is proposed:

- (1) Differentiated training for teachers of different age and experience levels;
- (2) Joint development of professionally adapted resources by universities, enterprises, and platform developers;
- (3) Structured guidance for self-regulated learning and process-based evaluation;
- (4) Strict data governance and technical protection protocols.

7. Conclusions and prospects

7.1. Conclusions

- (1) AI effectively resolves long-standing dilemmas in college mathematics teaching, including weak personalization, poor professional alignment, and lagged feedback.
- (2) The trinity framework of intelligent system, personalized resources, and blended teaching forms a closed-loop, high-effectiveness teaching model.
- (3) The hierarchical statistical framework ensures scientific and reliable conclusions.
- (4) Teacher capacity building and resource co-construction are decisive for successful implementation of AI-enabled teaching.

7.2. Prospects

Future applications of AI in college mathematics teaching will evolve toward deeper technological integration, finer personalization, closer cross-disciplinary linkage, and more diversified intelligent evaluation systems. Immersive scenarios, real-time adaptive push, disciplinary integration, and full-process intelligent assessment will become major developmental directions.

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